From Specification Shifts to Included Variable Bias in Dynamic Panels: A Diagnostic for Time-Varying Controls and a Benchmark for the Contemporaneous Treatment Effect

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Abstract

We study specification shifts that arise when researchers condition on observed covariates that may themselves respond to treatment. For linear nested models, the sample shift $\widehat{\Delta}_Z = \widehat{\beta}^{\star} - \widehat{\beta}$ has the exact decomposition $-\widehat{\theta}^{\star}\widehat{\pi}$. This algebra does not by itself establish causal bias: directed acyclic graphs, substantive timing, and identification assumptions determine whether the shift reflects collider bias, overcontrol, deconfounding, or an ambiguous change in projection. For time-series cross-sectional settings that target a linear, homogeneous contemporaneous treatment effect (CET), we evaluate ADL + FE with lagged state variables as a conditional baseline. In the reported collider, dual-role, and mediator designs, specifications with the relevant lagged state produce smaller CET bias than the displayed alternatives, while conditioning on a contemporaneous mediator moves the coefficient toward the direct effect. Applications illustrate the descriptive diagnostic and the timing judgments required before interpreting a shift causally. The baseline applies only with a defended causal clock and history, sequential exchangeability, adequate dynamics and support, no relevant interference, and a time dimension or bias correction appropriate for dynamic fixed-effects estimation.

1 Introduction

Researchers using time-series cross-sectional data constantly face a deceptively simple specification choice: should an observed time-varying covariate be included in the regression? Consider an international-relations

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scholar studying the effect of UN peacekeeping on democratization. Conditioning on conflict intensity is tempting because more severe conflicts are more likely to receive peacekeeping missions and are less likely to democratize. But peacekeeping can itself reduce violence. If so, conditioning on contemporaneous conflict intensity means conditioning on a variable that may already have responded to treatment. Omitting conflict may leave confounding; including realized conflict may introduce a different problem. The same logic appears whenever researchers want to control for GDP, aid, inflation, turnout, repression, or any other variable that evolves within the same period as treatment and outcome.

Recent work on difference-in-differences with time-varying covariates makes this problem stark. When a covariate can itself respond to treatment, both conditioning on the realized covariate and omitting it can be problematic, because the causally relevant object may be something like the untreated potential covariate $Z_t(0)$ rather than the realized Z_t (Caetano and Callaway, 2024; Caetano et al., 2022; Lin and Zhang, 2022). Related work on panels and causal dynamics likewise shows that time-varying covariates can create demanding identification problems (Blackwell and Glynn, 2018; Imai and Kim, 2021). We take that broader literature as the starting point rather than the target of critique.

Our contribution is different and narrower, but also more operational. We introduce the specification-shift diagnostic for cases in which researchers condition on observed covariates that may themselves respond to treatment. In linear nested models, the resulting shift admits a simple closed-form decomposition:

$$\widehat{\beta}^{\star} - \widehat{\beta} = -\widehat{\theta}^{\star}\widehat{\pi},$$

where $\widehat{\theta}^{\star}$ is the coefficient on the candidate control in the long regression and $\widehat{\pi}$ is the coefficient on treatment in an auxiliary regression for that control. The sample object $\widehat{\Delta}_Z = \widehat{\beta}^{\star} - \widehat{\beta}$ measures how much including Z moves the treatment coefficient. Directed acyclic graphs (DAGs), timing, and identification assumptions determine whether that descriptive shift can be interpreted as causal bias. This is the paper's general contribution. The current manuscript develops it for the linear models most common in TSCS work, where the decomposition is exact and directly estimable.

The paper's main applied payoff is narrower. For TSCS settings focused on the contemporaneous treatment effect (CET), we study ADL + FE with lagged state variables as a conditional baseline: lagged covariates enter only when their timestamps and causal roles are defended, while contemporaneous time-varying controls require an explicit causal clock. The coefficient can represent the CET only under the identification, functional-form, support, and estimation conditions stated below. The baseline is therefore a comparison specification for a defined TSCS problem, not a universal causal estimator or a rule that lagging a variable makes it admissible.

This architecture helps distinguish the paper from two nearby contributions. It is not only a warning that time-varying controls are hard, because the specification-shift decomposition gives that difficulty a named and estimable diagnostic. And it is not only a recommendation to use dynamic specifications, because the benchmark is presented as the main applied implication of a more general specification-shift framework. The broader research program behind this diagnostic may well travel beyond the current setting, but this paper develops its full payoff in the linear TSCS case.

We proceed in four steps. Section~2 defines the control-variable problem when observed covariates may themselves respond to treatment. Section~3 uses DAG logic to show why the same substantive variable can be harmful at t and useful at $t-1$. Section~4 presents the specification-shift diagnostic and derives the closed-form decomposition used in linear TSCS models. Section~5 then develops the main applied implication of that framework: a benchmark for the CET based on ADL + FE with lagged state variables. Section~6 shows how the diagnostic and the benchmark interact in published TSCS studies. Section~7 closes with the paper's claim, its limits, and the agenda it opens.

2 The Time-Varying Control Problem for the Contemporaneous Treatment Effect

Applied researchers working with time-series cross-sectional (TSCS) data rarely face a binary choice between `control'' and`do not control.’ ’ The more realistic problem is whether an observed time-varying covariate should enter the specification when the estimand is the contemporaneous treatment effect (CET), that is, the effect of D_t on Y_t. The same variable can help, hurt, or redefine the estimand depending on timing. A contemporaneous covariate Z_t may be a confounder, a collider, or a mediator. Its lag, Z_{t-1}, may instead behave as a predetermined state variable relative to the CET.

Standard heuristics are too coarse for this problem. `Control checking'' recommends adding predictors of the outcome to gain precision;`confounding checking’ ’ recommends adding joint causes of treatment and outcome (Cinelli, Forney and Pearl, 2022). Both heuristics ignore the possibility that a candidate control is post-treatment in the period of interest. That omission is costly in TSCS designs because treatment, outcome, and covariates evolve jointly over time, often with persistence in all three processes (Blackwell and Glynn, 2018; Imai and Kim, 2021).

Recent DID work sharpens the problem further: if Z_t may itself respond to treatment, neither conditioning on realized Z_t nor omitting it necessarily identifies the target estimand, because the relevant object may be the untreated potential covariate $Z_t(0)$ rather than the observed Z_t (Caetano and Callaway, 2024; Caetano et al., 2022). We do not resolve that identification problem in general. Instead, we ask a more operational question. When applied researchers condition on an observed time-varying covariate that may respond to treatment, how much does that choice move the treatment coefficient, and under what conditions can a dynamic TSCS specification serve as a CET baseline?

Our answer has two parts. The general contribution is an exact decomposition of the specification shift in linear nested models; calling that shift causal bias requires a DAG, a target estimand, and identification assumptions. The more specific contribution is the paper's payoff in TSCS settings with the CET: evaluating a dynamic baseline based on ADL + FE with lagged state variables under stated scope conditions. The formula quantifies the change between specifications, while the causal argument determines what that change means.

The scope qualification matters. A lagged control can be called pre-treatment only relative to the effect of D_t on Y_t, and only when its substantive timestamp supports that ordering. Even then, it can be treatment-responsive relative to cumulative, delayed, or long-run effects. The paper's conditional baseline is therefore explicitly local to the CET. Difference-in-differences and related panel designs remain part of the motivation for why time-varying controls are hard, but the paper's formal and applied results are developed in the linear TSCS setting.

3 DAGs, Timing, and Why Z_t Is Not Z_{t-1}

The causal role of a time-varying covariate depends on its substantive timing, not its subscript alone. Before classifying a control, we therefore need to map the variables recorded in a panel row onto a causal sequence within the period.

3.1 The CET and the Within-Period Causal Clock

Let H_{it}^- denote the information and state variables fixed immediately before the period-t treatment begins. For two well-defined treatment levels d and d', we define the contemporaneous treatment effect as

$$\tau_{\text{CET}}(d, d'; h) = \mathbb{E}\big[Y_{it}(d) - Y_{it}(d') \mid H_{it}^- = h\big] ,$$

where $Y_{it}(d)$ is measured after the period-t intervention and any within-period responses it induces. Thus, unless a controlled direct effect is explicitly targeted, the CET includes paths such as $D_{it} \to Z_{it} \to Y_{it}$. In a constant linear model with a scalar treatment, β_{CET} is the corresponding unit contrast, or marginal effect for a continuous dose.

The substantive clock underlying this definition is

$$H_{it}^{-} \longrightarrow D_{it} \longrightarrow Z_{it} \longrightarrow Y_{it}^{+},$$

where H_{it}^{-} is the state at the opening of the causal period, D_{it} records the onset and intensity of treatment, Z_{it} is allowed to respond, and Y_{it}^{+} is measured after the relevant exposure window. This ordering is an empirical claim, not a consequence of notation. Table~1 states the minimum timestamp information needed to assess it.

This displayed chain is the candidate clock when Z_t is a mediator. It is not imposed on every graph below. A contemporaneous-collider claim $D_t \to Z_t \leftarrow Y_t$ instead requires the outcome process represented by Y_t to precede the measurement of Z_t within the causal period, while D_t can affect both. Thus the common subscript denotes a panel period, not simultaneity or a universal within-period order; each application must defend the mechanism-specific clock.

Table 1: Minimum timestamp record for defining the contemporaneous treatment effect. A lag label is not evidence that a covariate is pretreatment.

Variable	Minimum timestamp record	Requirement for the CET clock
D	Onset, end, and intensity window; whether the measure is a stock, flow, event, or period average	Defines the intervention or exposure contrast within period t
Z	Reference and measurement window; whether it can respond to anticipated, prior, or current treatment	Must close before the first relevant exposure to be called pretreatment; otherwise classify its timing as plausible or ambiguous
Y	Reference window and measurement time; whether any part is recorded before treatment begins	Must follow the relevant exposure window for the stated CET; overlapping aggregates require a finer estimand

Three complications are common in annual and other aggregated panels. First, anticipation can make a recorded Z_{t-1} treatment-responsive before the formal onset of D_t. Second, when treatment varies continuously within the period, a scalar annual value may summarize a treatment path with repeated $D \leftrightarrow Z \leftrightarrow Y$ feedback; the relevant intervention is then a contrast between exposure trajectories, not automatically a one-unit change in the annual average. Third, annual measures may overlap: a late-year treatment can occur after much of Z_t or Y_t has already been realized, while a beginning-of-year outcome may precede the nominally contemporaneous treatment. When documentation establishes nonoverlapping windows, timing is known; when institutional knowledge supports but does not timestamp the sequence, it is plausible; and when windows overlap or remain undocumented, it is ambiguous. The lagged-control benchmark below applies only after this clock has been defended.

3.2 Dynamic Causal Structures for the CET

The elementary fork, chain, and collider remain the building blocks of causal graphs (Pearl, 2009; Elwert and Winship, 2014), but TSCS specifications combine those structures across periods. Figure 1 makes the inherited path explicit for a contemporaneous collider. The blue arrows mark the treatment–outcome causal edges in successive periods. Conditioning on Z_t opens $D_t \to Z_t \leftarrow Y_t$. Because Z_t is also a descendant of the lagged collider Z_{t-1}, that conditioning can activate the inherited path $D_t \leftarrow D_{t-1} \to Z_{t-1} \leftarrow Y_{t-1} \to Y_t$. Conditioning on Y_{t-1} blocks this inherited path at a noncollider, but it does not close the contemporaneous path opened at Z_t. The graph also shows the ordinary dynamic path $D_t \leftarrow D_{t-1} \to Y_{t-1} \to Y_t$; that path is distinct from collider activation and is likewise blocked at Y_{t-1}.

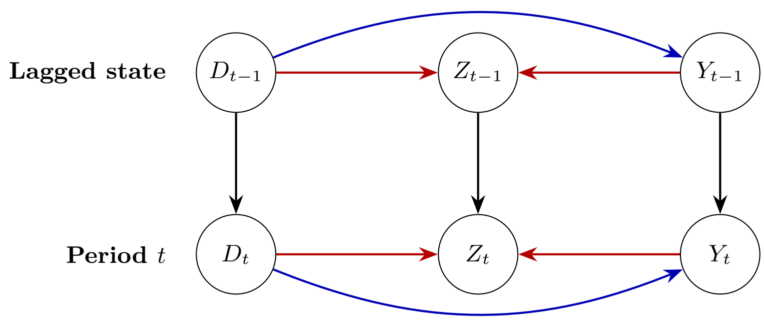

Figure 1: Dynamic contemporaneous-collider DAG for the CET. Red arrows meet at Z_t and Z_{t-1}; blue marks the causal $D_t \to Y_t$ path. Conditioning on Z_t opens the contemporaneous collider and, as conditioning on a descendant of Z_{t-1}, can also activate the inherited lag path.

Figure 2 represents a different role for the same contemporaneous variable. Here Z_t transmits part of the effect of D_t to Y_t, while Z_{t-1} is a lagged state that can affect current treatment and outcome. For the total CET, both current-period blue paths belong to the estimand. Conditioning on a substantively pretreatment Z_{t-1} blocks $D_t \leftarrow Z_{t-1} \to Y_t$ without blocking the within-period mediated path. The remaining lag path $D_t \leftarrow D_{t-1} \to Y_{t-1} \to Y_t$ requires D_{t-1}, Y_{t-1}, or other sufficient history. Conditioning on Z_t instead blocks $D_t \to Z_t \to Y_t$ and changes the target to a direct-effect parameter under additional mediation assumptions.

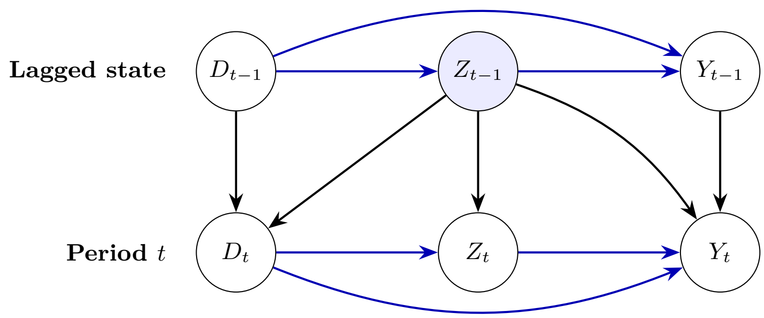

Figure 2: Dynamic contemporaneous-mediator DAG with lagged confounding and state dependence. For the total CET, the direct path and the path through Z_t remain open. Conditioning on a genuinely pretreatment Z_{t-1} can block lagged confounding; conditioning on Z_t blocks the displayed mediated path.

3.3 Timing Logic for Candidate Controls

Table~2 summarizes the main timing logic. The key point is that the paper's benchmark is anchored to the CET rather than to a generic slogan such as “lag everything.’ ’

Table 2: Timing logic for observed controls when the estimand is the contemporaneous treatment effect. Subscripts identify candidate roles; the substantive clock determines the causal classification.

Candidate control	Possible role for the CET	Main risk or benefit	Conditional action
Z_t	Collider if post-treatment	Opens a noncausal collider path	Do not condition unless the DAG and causal clock rule out post-treatment status
Z_t	Mediator if post-treatment	Blocks part of the total effect	Do not condition if the target is the total contemporaneous effect
Z_{t-1}	Predetermined state if timestamp passes	Captures state dependence	Candidate baseline control only after the timestamp check
Z_{t-1}	Lagged confounder if timestamp passes	Removes OVB without blocking the contemporaneous path	Candidate baseline control only after the timestamp check

Three cases are especially relevant for the remainder of the paper. First, conditioning on a contemporaneous collider Z_t opens a noncausal collider path and can create included variable bias relative to the CET when the graph and identification assumptions establish that interpretation. Second, a contemporaneous mediator Z_t creates over-control if the target is the total contemporaneous effect. Third, a dual-role variable can be both a lagged confounder and a contemporaneous collider. In that case, the empirical question is not whether the variable should ever appear in the specification, but whether its documented measurement window places it before or after the relevant treatment begins.

4 The Specification-Shift Diagnostic

The paper's general contribution is to define the specification shift as a distinct diagnostic object. Once researchers accept that an observed control may itself respond to treatment, they need a language for the resulting coefficient change. The decomposition quantifies that change in a given linear specification; the DAG, timing, estimand, and identification assumptions determine its causal meaning. We reserve the term included variable bias (IVB) for cases in which those elements establish that including Z moves the estimator away from the contemporaneous-effect target β_{CET}.

Consider the short model

$$Y = \beta D + W\delta + u \tag{1}$$

and the long model

$$Y = \beta^{\star} D + \theta^{\star} Z + W\delta^{\star} + u^{\star}, \tag{2}$$

where W collects admissible controls. By the Frisch–Waugh–Lovell theorem,

$$\widehat{\beta}^{\star} - \widehat{\beta} = -\widehat{\theta}^{\star}\widehat{\pi}, \tag{3}$$

where $\widehat{\pi}$ is the coefficient on D in the auxiliary regression of Z on D and W.

Proposition 1 (Multiple candidate controls)**.** *Under the usual full-rank conditions for unweighted OLS and using the same observations and common regressors, let $Z = (Z_1, \ldots, Z_q)'$, let $\widehat{\theta}$ be its coefficient vector in the joint long regression, and let $\widehat{\pi}$ collect the coefficients on D from the auxiliary regressions of each Z_j on D and W. Then*

$$\widehat{\Delta}_{\mathcal{Z}} = \widehat{\beta}_{\mathcal{Z}} - \widehat{\beta}_{\varnothing} = -\widehat{\theta}'\widehat{\pi}.$$

For any order of inclusion, the conditional increments telescope to $\widehat{\Delta}_{\mathcal{Z}}$, but their control-specific allocation generally depends on that order. Leave-one-control-out increments are instead last-in conditional comparisons and need not sum to the joint shift.

The vector identity remains a diagnostic for nested linear TSCS specifications, not an automatic decomposition of causal bias. Correlated controls make the distinction especially important: the endpoint shift is unique, whereas sequential allocations are path dependent and leave-one-out comparisons answer a different sensitivity question. The online appendix gives the FWL proof, formal definitions, and a deterministic correlated-control example.

For reporting, we recommend the endpoint shift together with leave-one-control-out comparisons, each defined as the full-model coefficient minus the coefficient from a model omitting one candidate control while retaining the others. These are conditional sensitivity checks, not additive shares of the joint shift. A symmetric Shapley allocation can summarize the average contribution across all inclusion orders when the candidate set is small, but it remains an allocation of a linear projection movement; we therefore place it in the online appendix rather than treat it as a causal decomposition.

In cross-sections, W may contain baseline covariates. In TSCS settings, W can include fixed effects, lagged dependent variables, lagged treatment, and other lagged state variables. The formula is therefore an exact linear-model identity, not a separate estimator. That exactness makes the specification-shift diagnostic directly readable from nested linear regressions already familiar to applied researchers.

Two interpretive points matter. First, the formula is diagnostic rather than self-justifying. If Z is a confounder, the same arithmetic difference reflects deconfounding rather than harm. If Z is a mediator, the difference reflects movement from the total to the direct effect (Acharya, Blackwell and Sen, 2016). Only the DAG tells the researcher which interpretation applies. Second, the formula remains useful precisely because it converts a vague modeling dispute into two estimable components: how strongly Z predicts the outcome in the long model, and how strongly D predicts Z once admissible controls have been partialled out.

This is also where the paper's originality lies. Work on dynamic identification, sequential ignorability, and DID with time-varying covariates shows why observed controls may be problematic (Blackwell and Glynn, 2018; Imai and Kim, 2021; Caetano and Callaway, 2024). The paper contributes a different object: a named and estimable decomposition of the specification shift itself. It develops that object in the linear TSCS domain, where it yields both a diagnostic and, in the next section, a conditional comparison baseline. Whether analogous specification-shift diagnostics can be built for synthetic-control-style estimators, synthetic DID, or factor models is an important agenda, but not a result claimed here.

For the CET in TSCS settings, the practically relevant auxiliary regression is dynamic. If the benchmark specification is

$$Y_{it} = \beta D_{it} + \rho Y_{i,t-1} + \kappa D_{i,t-1} + X_{i,t-1}'\lambda + \alpha_i + \tau_t + \varepsilon_{it}, \tag{4}$$

then the corresponding decomposition asks how much the treatment coefficient shifts when a candidate control is moved from the lagged-state set $X_{i,t-1}$ into the contemporaneous set. This is the operational meaning of diagnosing a candidate time-varying control in the remainder of the paper.

The formula also clarifies a point that is easy to miss in practice: lagging the treatment does not mechanically remove the problematic path. If D_t is autocorrelated, a collider that is contemporaneous with D_t can remain associated with D_{t-1}. What matters is not whether the treatment index moved backward by one period, but whether the specification blocks the relevant noncausal path for the estimand of interest.

4.1 Conditions Under Which the ADL + FE Coefficient Represents the CET

The ADL + FE coefficient represents the CET only under explicit scope conditions. Let H_{it}^{-} be the observed pre-exposure history defined above. A calendar-lag vector $H_{i,t-1}$ is a valid implementation only when the timestamp checklist establishes $H_{i,t-1} \subseteq H_{it}^{-}$ and the included lags and transformations are sufficient for current assignment and the conditional outcome mean. A lag label alone establishes neither condition. A sufficient constant-coefficient restriction is

$$\mathbb{E}[Y_{it}(d) \mid H_{it}^{-}, \alpha_i, \tau_t] = \alpha_i + \tau_t + r(H_{it}^{-})'\gamma + \beta_{\text{CET}} d,$$

together with consistency, no interference, no anticipation, sequential mean exchangeability of D_{it} given H_{it}^{-}, and conditional positivity for the treatment contrast. For discrete treatment, the target levels must have positive conditional probability; for continuous treatment, the target doses must lie in the conditional common support with local support around them. The causal clock must also be recursive: D_{it} precedes the contemporaneous outcome innovation, so simultaneous $Y_{it} \to D_{it}$ feedback is excluded. By contrast, $Y_{i,t-1} \to D_{it}$ feedback is compatible with identification when $Y_{i,t-1} \in H_{it}^{-}$ and the conditioned history is sufficient.

These causal support conditions are distinct from the rank condition for the population projection. If $\widetilde{D}_{it}$ is the population residual from projecting D_{it} on the included history and the unit and time effects, and $0 < \mathbb{E}[\widetilde{D}_{it}^2] < \infty$, then

$$\beta_D^P = \frac{\mathbb{E}[\widetilde{D}_{it} Y_{it}]}{\mathbb{E}[\widetilde{D}_{it}^2]} = \beta_{\text{CET}}.$$

The online appendix states this result as a proposition and gives the full proof. This equality is a population identification result. Unit effects absorb additive time-invariant unit factors and time effects absorb additive common shocks; neither removes omitted time-varying confounding or heterogeneous responses to common shocks. A misspecified lag order or nonlinear history also breaks the stated conditional mean. If treatment effects or dynamics vary over units, periods, or histories, the single coefficient is generally a projection-weighted summary rather than a common CET.

Table 3: Standardized scope conditions for using ADL + FE with lagged state variables as a CET baseline.

Condition	Required scope condition	If unmet
CET target and causal clock	A well-defined CET; consistency; no relevant spillovers or interference; defended within-period timing; no anticipation or simultaneous $Y_{it} \to D_{it}$ feedback	The stated intervention, ordering, or unit-level causal contrast is not identified
Observed history and exchangeability	$H_{i,t-1} \subseteq H_{it}^{-}$ and sufficient observed history; sequential mean exchangeability; no unobserved contemporaneous cause of D_{it} and Y_{it}	The coefficient can retain confounding or condition on a treatment-responsive state
Dynamic conditional mean	Plausible lag order and transformations that span assignment and the dynamic conditional outcome mean	The regression can omit relevant dynamics even when every included variable is correctly timed
Functional form and effect summary	An additive linear response and a homogeneous CET if one coefficient is to represent a common effect	The coefficient is generally a projection-weighted or best-linear-approximation parameter
Fixed effects and common shocks	Additive unit heterogeneity and common time shocks; no omitted treatment-correlated time-varying unit process or heterogeneous common-shock response	Fixed effects do not absorb the omitted process, so population identification can fail
Support and projection rank	Conditional overlap for the target contrast and $0 < \mathbb{E}[\widetilde{D}_{it}^2] < \infty$ after partialling out history and fixed effects	The contrast extrapolates, or the population FWL coefficient is undefined
Finite-T estimation	A time dimension adequate for within-estimator bias, or a bias correction or alternative estimator whose assumptions fit the panel	The population CET can be identified while the conventional within estimate has Nickell-type bias
Inference	An uncertainty procedure appropriate for serial and cross-sectional dependence	Point identification can be unchanged while standard errors and confidence intervals are invalid

The distinction between population identification and estimation matters in dynamic panels. With lagged outcomes, demeaning generally correlates the transformed lag with the transformed innovation, producing Nickell-type bias at finite T even when the level conditional mean identifies β_{CET} (Nickell, 1981). Serial and cross-sectional dependence are separately inference problems when the remaining innovations are conditionally mean zero; an omitted common factor correlated with treatment is instead an identification problem. For short panels, bias corrections or Arellano–Bond may be useful alternatives under their own moment and instrument conditions, but GMM is not uniformly superior and does not repair a failed causal clock (Arellano and Bond, 1991). These qualifications align the benchmark with the sequential-identification logic in the dynamic-panel literature (Blackwell and Glynn, 2018; Imai and Kim, 2021).

5 Benchmarking Dynamic Specifications for the CET

This section develops the paper's main applied implication of the specification-shift framework. Conditional on Table~3, an informative applied comparison for the CET is between dynamic specifications that differ in whether they condition on contemporaneous Z_t, rather than treating a static TWFE short regression as a causal reference. The baseline does not substitute for the specification-shift diagnostic: the former defines a conditional comparison specification, while the latter measures the coefficient change.

5.1 Collider and Dual-Role Controls in TSCS

We begin with the dual-role setting in which Z_{t-1} is a confounder for (D_t, Y_t), but Z_t is contemporaneously affected by both treatment and outcome. In this structure, a variable such as GDP can serve as a lagged state variable while its contemporaneous value has a different causal role.

The dual role is visible in Figure 3. Without conditioning, $D_t \leftarrow Z_{t-1} \rightarrow Y_t$ is an open backdoor path. Conditioning on Z_{t-1} closes that fork but opens the inherited path $D_t \leftarrow D_{t-1} \rightarrow Z_{t-1} \leftarrow Y_{t-1} \rightarrow Y_t$, because Z_{t-1} is a collider on that path. Adding Y_{t-1} blocks this inherited path at the noncollider Y_{t-1} and also blocks the distinct omitted-dynamics path $D_t \leftarrow D_{t-1} \rightarrow Y_{t-1} \rightarrow Y_t$. This d-separation statement is conditional on the displayed graph: it does not close $D_t \rightarrow Z_t \leftarrow Y_t$ if Z_t is conditioned on, and it does not rule out other omitted paths or finite-T estimation bias.

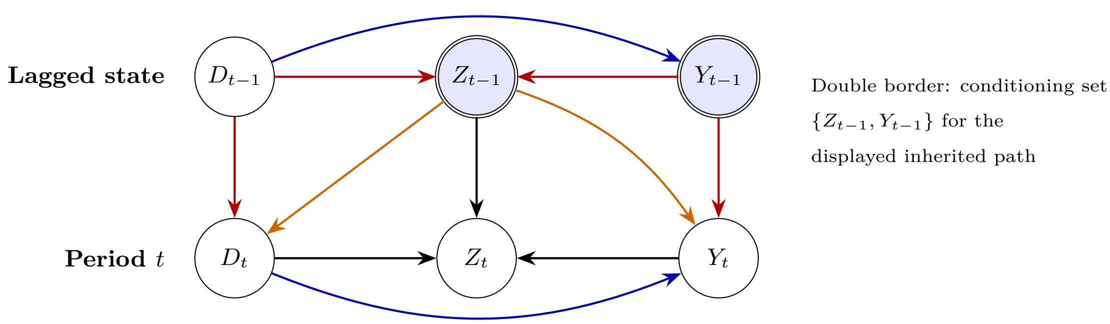

Figure 3: Dynamic dual-role DAG. Orange arrows form the lagged-confounding fork; red arrows trace the inherited collider path. Conditioning on Z_{t-1} blocks the former but opens the latter. Conditioning additionally on Y_{t-1} blocks that inherited path, subject to the displayed DAG and the paper's timing and identification conditions.

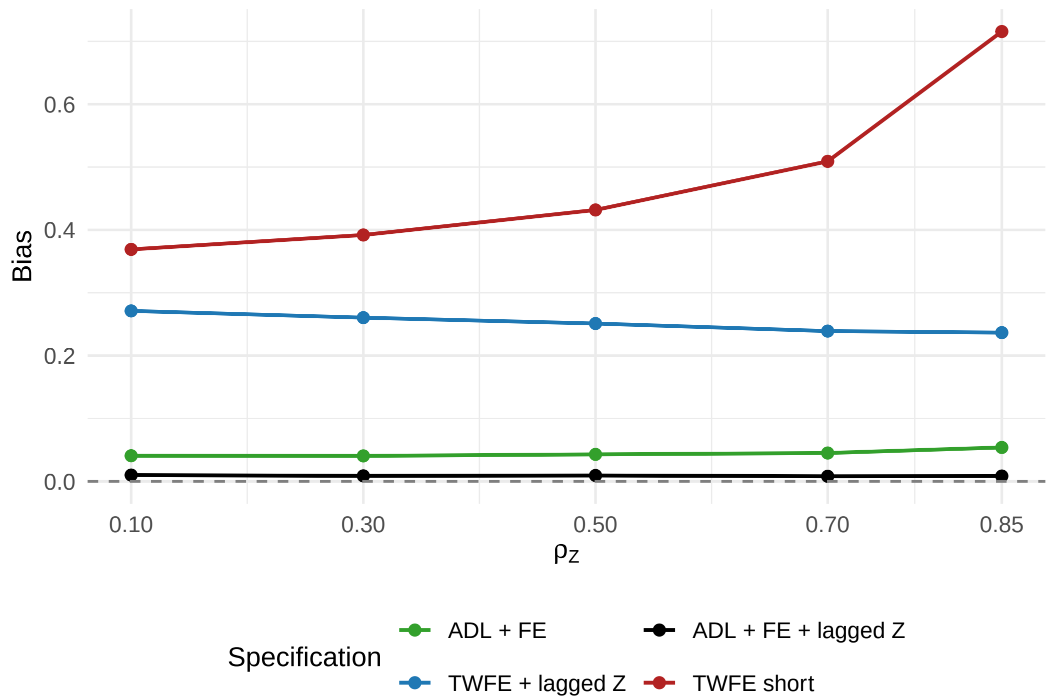

Figure 4: Dual-role simulation comparing static and dynamic specifications. In the main design, ADL + FE + Z_{t-1} keeps bias near zero across the reported range of collider persistence, while the static TWFE specifications retain appreciable bias.

In the main dual-role design, the bias of ADL + FE + Z_{t-1} stays between 0.008 and 0.010, while the bias of TWFE + Z_{t-1} remains between 0.237 and 0.271. The static short model has the largest reported bias, reaching 0.716 in the most persistent scenarios. For this DGP, the relevant comparison is therefore between specifications that represent Z as a lagged state inside the dynamic outcome model and specifications that omit part of that state.

In the displayed DGP, outcome persistence transmits both the inherited collider path and the ordinary lagged treatment–outcome path through Y_{t-1}. Table 4 shows that when this edge is removed by setting $\rho_Y = 0$, the TWFE bias in the reported design is approximately zero. Conditioning on Y_{t-1} blocks both displayed paths when $\rho_Y > 0$, subject to the graph and scope conditions stated above; the simulation contrast does not identify how much of the improvement comes from each path.

This finding disciplines the paper's recommendation. The lagged outcome is not generally sufficient. In the displayed DGP, it conditions on the persistent outcome state through which both cited lag paths reach Y_t. ADL + FE with Y_{t-1} and substantively justified lags of candidate controls is therefore the comparison baseline for this dual-role design, conditional on the identification and estimation conditions above.

Table 4: Bias by outcome persistence in the dual-role DGP. When $\rho_Y = 0$, the reported bias from conditioning on Z_{t-1} is approximately zero even without Y_{t-1}. When $\rho_Y > 0$, including Y_{t-1} conditions on the displayed persistent outcome state. This comparison does not decompose the ordinary dynamic and inherited-collider paths.

Outcome persistence (ρ_Y)	TWFE + Z_{t-1}	ADL + FE + Z_{t-1}
0.0	0.001	0.008
0.5	0.251	0.009
0.8	0.261	0.004

5.2 Over-Control with Contemporaneous Mediators

The second recurring problem is not collider bias but over-control. When Z_t lies on the path from D_t to Y_t, conditioning on Z_t shifts the estimate toward the direct effect. For the CET, the reported comparison is dynamic: an ADL specification that leaves the contemporaneous mediator out of the regression, one that conditions on Z_t, and one that includes Z_{t-1}. Whether the lagged version belongs in the baseline depends on the timing and adjustment conditions above.

Table 5: Over-control simulations for the CET. In both the pure-mediator and mediator-plus-confounder designs, the displayed ADL specification with contemporaneous Z_t produces a coefficient near the direct-effect target, whereas the displayed specification with Z_{t-1} produces a coefficient near the total-effect target. These are design-specific comparisons.

Scenario	Target total	Target direct	ADL, no Z	ADL + Z_t	ADL + Z_{t-1}
Pure mediator, $\rho_Z = 0.3$	1.20	1	1.201	0.998	1.199
Pure mediator, $\rho_Z = 0.7$	1.20	1	1.199	0.992	1.197
Mediator + confounder, $\rho_Z = 0.5$	1.03	1	1.091	1.023	1.031
Mediator + confounder, $\rho_Z = 0.7$	1.03	1	1.111	1.016	1.028

The clean mediator design is the simplest illustration. The ADL specification without Z estimates about 1.20, the ADL specification with Z_t estimates about 1.00, and the ADL specification with Z_{t-1} remains near 1.20. In this DGP, those values correspond respectively to the total-effect target, the direct-effect target, and the total-effect target. In the mediator-plus-confounder design, including Z_{t-1} removes the displayed lagged backdoor path without blocking $D_t \to Z_t \to Y_t$; that interpretation depends on the known DGP and does not follow from the specification label alone.

For this reason, the static TWFE short specification is not the primary comparison for the mediator DGP. Without the relevant dynamics, its coefficient mixes the effect of conditioning on Z_t with lag misspecification. The reported operational contrast is therefore ADL without Z, ADL with Z_t, and ADL with Z_{t-1}.

5.3 Conditional Workflow for the CET Baseline

For applied TSCS work centered on the CET, the baseline is usable only through the following conditional workflow:

1. Define the CET and document the exposure, covariate, and outcome windows; do not infer causal order from panel subscripts.
2. Assess every condition in Table~3, including history sufficiency, exchangeability, lag structure, support, finite-T behavior, and interference.
3. If those conditions are defended, use ADL + FE with the required lagged state variables as a baseline and state the exact lag order and transformations.
4. Compare that baseline with the specification containing a candidate contemporaneous Z_t and report the resulting specification shift $\widehat{\Delta}_Z$.
5. Interpret $\widehat{\Delta}_Z$ as causal bias only when the DAG, timing, and identifying assumptions establish the CET interpretation; otherwise retain the descriptive specification-shift language.
6. Choose another target and method when the estimand is distributed, cumulative, long-run, or a treatment regime rather than the CET.

Additional simulations delimit this recommendation. When the reported outcome equations contain the tested nonlinear effects of Z_{t-1} or level-dependent trends, the absolute bias of ADL(all) does not exceed 0.4 percent of β across the stable scenarios in that grid. This result describes those scenarios; it does not establish adequacy under untested nonlinearities or lag structures. In the contemporaneous-confounding simulations, the bias of ADL(all) instead ranges from 8 to 53 percent of β, while the oracle specification that observes the confounder keeps bias below 0.3 percent. The baseline therefore does not address unobserved contemporaneous confounding.

This baseline is intentionally narrower than a recommendation for panel estimators in general. It is one conditional comparison specification for observed time-varying controls when the target is the CET and the scope conditions in Table~3 are defended.

6 Applications

The applications serve two purposes. First, they show that the decomposition yields empirical specification-shift diagnostics rather than only theoretical algebra. Second, they show why those shifts require separate timing and causal judgments. Across the 14 theory-selected candidate controls in these six studies, the median $|\widehat{\Delta}_Z|/\text{SE}$ is 0.13, and 1 case exceeds one standard error. These are descriptive results from a purposive set of applications, not estimates of the prevalence or magnitude of causal bias in TSCS research.

6.1 Application 1: Democracy and Ethnic Inequality

Leipziger (2024) estimates the effect of democratization on ethnic inequality with a two-way fixed-effects design in which democracy and GDP per capita are both lagged one calendar year relative to the outcome. Indexing by the outcome year, the estimated exposure contrast is therefore $D_{t-1} \to Y_t$; equivalently, if $s = t-1$ denotes the exposure year, it is $D_s \to Y_{s+1}$. This application is an audit of a lagged exposure contrast, not a direct demonstration of the manuscript's core contemporaneous $D_t \to Y_t$ CET. The comparison remains useful because the same timestamp questions determine whether GDP closes before the democracy exposure in year s and because the coefficient shift shows how sensitive the lagged contrast is to conditioning on GDP. GDP is pre-outcome, but not demonstrably pretreatment relative to democracy: the two predictors share the same annual index, and the replication documentation does not provide within-year transition and measurement dates. We therefore classify the timing of GDP as ambiguous for the treatment contrast, rather than treating its lag label as dispositive. The practical question is whether GDP should be treated as a confounder, a treatment-responsive variable, or a dual-role variable.

GDP per capita is a plausible dual-role variable in this setting. Democratization can raise income over time (Acemoglu et al., 2019; Papaioannou and Siourounis, 2008), while ethnic inequality can depress income through weaker public-goods provision and conflict channels (Grundler and Link, 2024). The decomposition therefore does not mechanically tell us to drop GDP. It reports how the treatment coefficient changes when GDP enters the specification; the ambiguous timing prevents that shift from receiving a unique causal interpretation.

Table 6: Specification-shift decomposition for Leipziger (2024), public-services inequality measure.

Component	Estimate
$\hat{\beta}_{\text{short}}$	-0.0407
$\hat{\beta}_{\text{long}}$	-0.0352
$\hat{\theta}^{\star}$	-0.0497
$\hat{\pi}$	0.1123
$\widehat{\Delta}_Z = -\hat{\theta}^{\star}\hat{\pi}$	0.0056
$\lvert\widehat{\Delta}_Z\rvert/\text{SE}$	0.48
$\lvert\widehat{\Delta}_Z\rvert/\lvert\hat{\beta}_{\text{long}}\rvert$	15.9%

The estimated specification shift is 0.0056, or about 0.48 standard errors. The coefficient in the specification with GDP is roughly 16 percent smaller in absolute shift terms, but the difference remains below one reported standard error. Because GDP's causal role and within-year timing are ambiguous, this comparison does not establish attenuation bias or deconfounding.

6.2 Application 2: Postal Infrastructure and Economic Growth

Rogowski et al. (2022) provide the complementary case. Their panel is organized in five-year steps, the outcome is led one panel period, and the postal-stock and GDP measures enter the same regression row. Letting s index the baseline five-year row, the estimated exposure contrast relates the cumulative postal stock D_s to forward growth Y_{s+1}, where $s+1$ is the next five-year panel step. This application is therefore an audit of a forward exposure contrast, not a direct demonstration of the manuscript's core contemporaneous $D_t \to Y_t$ CET. It remains useful for timing and sensitivity because a cumulative stock makes prior exposure explicit and because the coefficient shift reveals how strongly the forward contrast depends on conditioning on baseline GDP. Baseline GDP is a plausible state variable for subsequent growth, but it is not demonstrably prior to the treatment represented by an accumulated stock of post offices; it may already embody responses to earlier postal investment. Its timing is ambiguous for the treatment contrast even though it precedes the measured outcome window. Here the specification-shift decomposition reveals a large arithmetic shift but cannot decide, by itself, whether dropping GDP improves identification.

Table 7: Specification-shift decomposition for Rogowski et al. (2022).

Control as z	$\hat{\beta}_{\text{short}}$	$\hat{\beta}_{\text{long}}$	$\hat{\theta}^{\star}$	$\hat{\pi}$	$\widehat{\Delta}_Z$	$\lvert\widehat{\Delta}_Z\rvert$/SE	$\widehat{\Delta}_Z/\hat{\beta}$ (%)
Log GDP p.c.	0.0083	0.0198	-4.0130	0.0029	0.0115	2.11	58.0
Log population	0.0206	0.0198	-1.9465	-0.0004	-0.0008	0.15	-4.0
Urbanization	0.0192	0.0198	-3.1421	0.0002	0.0006	0.11	3.0
Polity2	0.0191	0.0198	0.0341	-0.0221	0.0008	0.14	3.8

For lagged GDP per capita, $|\widehat{\Delta}_Z|/\text{SE} = 2.11$, the largest value among the published applications. But this is precisely the case in which the formula must not be overread. GDP's causal role is unresolved: conditional convergence makes it a possible confounder; inherited postal infrastructure may make it a legacy treatment-responsive state or mediator; and both channels could give it a dual role. The large coefficient shift is therefore descriptive until a DAG and evidence on the relevant measurement and exposure windows establish more. The lesson is not "drop GDP.' ' The lesson is that a large specification shift demands explicit causal reasoning about timing and role. This is the use case for a diagnostic paired with a conditional baseline rather than an automatic decision rule.

7 Conclusion

This paper studies specification shifts that arise when researchers condition on observed covariates that may themselves respond to treatment. In the linear nested models used throughout TSCS work, $\widehat{\Delta}_Z$ admits a simple closed-form decomposition. That identity does not decide whether a control is admissible and does not, by itself, measure causal bias. It quantifies the size and direction of the coefficient shift; a CET, a time-indexed DAG, defended timing, and identification assumptions are required for causal interpretation.

The paper's main applied payoff is narrower and more concrete. For TSCS settings focused on the CET, ADL + FE with lagged state variables is a conditional baseline only when the target and causal clock are defined, the observed history is sufficient, sequential exchangeability and support are credible, the dynamic conditional mean is plausible, relevant interference is absent, and finite-T bias is negligible or addressed. The simulations show how that specification compares with the displayed alternatives in the reported dual-role, inherited-path, and mediator DGPs; they do not establish it as a causal estimator outside those conditions.

This combination of a descriptive diagnostic and a conditional TSCS baseline is the paper's contribution. The diagnostic reports how coefficients move across nested specifications; the baseline defines a structured comparison when the CET scope conditions hold. Keeping those roles separate prevents the observed specification shift from being treated as causal bias without the required assumptions.

The limits are equally important. The baseline is local to the CET. A lagged covariate that is pre-treatment for $D_t \to Y_t$ can be treatment-responsive for distributed, cumulative, or long-run effects. The reported nonlinearity results cover only the tested stable scenarios, while the contemporaneous-confounding designs show substantial remaining bias when a current common cause is unobserved. This paper also does not claim results for synthetic-control-style estimators, synthetic DID, factor models, treatment regimes, or nonlinear links. The present contribution is limited to the specification-shift diagnostic and its conditional use in linear TSCS settings.

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